

A

PBL Report on

Hydraulic Braking System

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IN

ELECTRICAL AND ELECTRONICS ENGINEERING

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His mentorship has not only enriched our technical understanding but also inspired me to approach engineering challenges with curiosity and creativity.

ABSTRACT

This project explores the design, implementation, and characterization of an affordable and practical precision angular positioning system utilizing a 28BYJ-48 unipolar stepper motor driven by a ULN2003 motor driver module, controlled by an Arduino microcontroller. Stepper motors are fundamental electromechanical devices vital in applications demanding accurate and repeatable rotational movements, such as small-scale robotics, automated indexing, and consumer electronics.

The primary objective of this project is to provide a hands-on understanding of unipolar stepper motor operation and its open-loop control for achieving precise angular displacements. The system utilizes the readily available and cost-effective 28BYJ-48 stepper motor, known for its internal gearbox providing high resolution, coupled with the ULN2003 Darlington array IC, which simplifies the interfacing between the Arduino and the motor coils. The Arduino Uno microcontroller is programmed to generate the appropriate phase sequences (full-step or half-step) to energize the motor coils, thereby dictating the motor's direction and the exact number of steps it takes.

Through the practical construction and experimentation with this setup, this project aims to demonstrate:

Unipolar Stepper Motor Fundamentals: The principles of coil energization for unipolar motors, the effect of the internal gearbox on resolution, and the differences between full-step and half-step sequencing.

Arduino-Based Control: The process of writing code to generate precise pulse patterns for accurate angular positioning.

Open-Loop Control Characteristics: Evaluation of the system's accuracy, repeatability, and limitations inherent in open-loop stepper motor control.

Introduction

In modern engineering applications, precision and control are crucial factors in the automation of mechanical systems. One such application is **angular positioning**, where the rotation of a mechanical element must be controlled with high accuracy. This is particularly important in fields such as robotics, CNC machining, antenna positioning, and camera gimbal systems.

This project focuses on achieving **precision angular positioning** using a **stepper motor controlled by an Arduino microcontroller**. Stepper motors are widely used for such applications because of their ability to divide a full rotation into a large number of discrete steps, allowing for highly accurate control of angular motion without the need for feedback systems in basic implementations.

The core idea of the project is to use an Arduino to send precise electrical pulses to the stepper motor driver, which in turn controls the rotation of the stepper motor. By programming the Arduino with the desired angle, speed, and direction of rotation, we can achieve controlled and repeatable angular positioning.

This project demonstrates how an inexpensive and easily programmable microcontroller like Arduino, in combination with a stepper motor, can be used to design a cost-effective and efficient angular positioning system.

WORKING

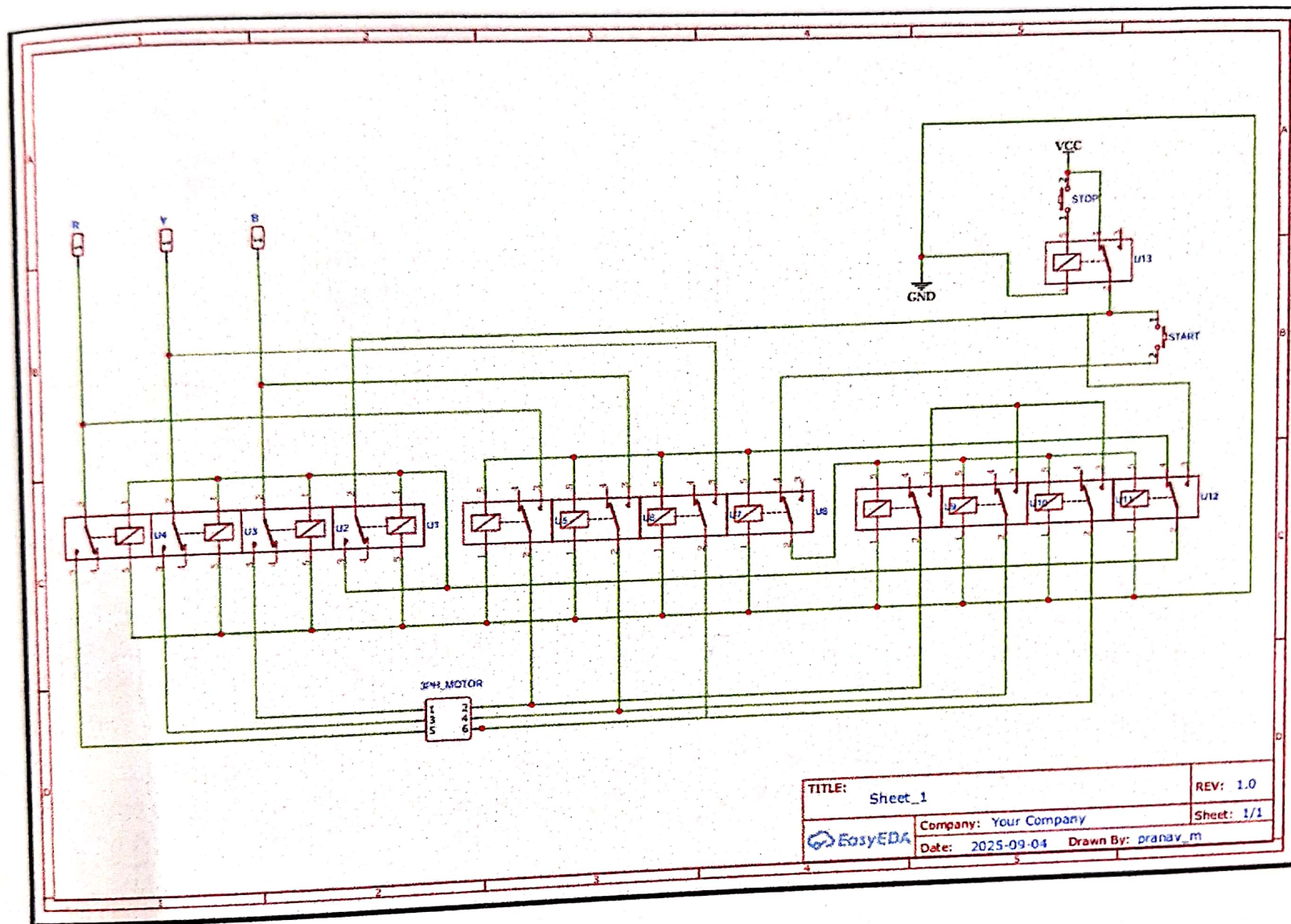
COMPONENTS USED

- **Arduino Uno:** Acts as the central controller that sends step signals to the motor driver.
- **Stepper Motor (28BYJ-48):** Converts electrical pulses into mechanical rotation in discrete steps.
- **Stepper Motor Driver (ULN2003):** Interprets step and direction signals from the Arduino and controls power delivery to the motor coils.
- **Power Supply:** Provides the required voltage and current for the motor.
- **Rotary Encoder or Potentiometer (Optional):** For real-time angle feedback or manual angle setting.

WORKING PRINCIPLE

1. **Command Input:** The user inputs the desired angular position via serial monitor, potentiometer, or pre-programmed values in the Arduino code.
2. **Signal Generation:** The Arduino calculates the number of steps needed to achieve the desired angular displacement. For example, if the motor takes 200 steps per revolution (1.8° per step), rotating 90° would require 50 steps.
3. **Motor Control:** The Arduino sends pulses to the stepper driver. The driver energizes the motor coils in the required sequence, causing the motor to rotate step by step.
4. **Precision Movement:** Because each step represents a fixed angular movement, the final angular position is highly predictable. Microstepping can be enabled on the driver to further increase precision.
5. **Optional Feedback (Advanced):** In more advanced versions, a rotary encoder could be used to verify the final position and correct any missed steps, though basic stepper systems are typically open-loop.

Circuit Diagram:



Advantages:

1. Reduced Starting Current
2. Reduced Electrical Stress
3. Lower Mechanical Stress
4. Energy Efficient During Start
5. Cost-Effective
6. Reliable and Simple Design
7. Good for Motors with Light Starting Loads

Limitations:

1. Not suitable for high starting torque loads (like conveyors or compressors).
2. Sudden torque jump during transition can cause mechanical stress.
3. Not ideal for motors that start and stop frequently.

Applications:

1. Pumps (centrifugal, submersible)
2. Fans and blowers
3. Conveyors
4. Air compressors
5. Crushers
6. Industrial mixers
7. Large machine tools
8. Marine applications (ship motors)
9. Textile machinery
10. HVAC systems (large chillers and compressors).

Applications

The system developed in this project can be applied to a variety of real-world scenarios where accurate angular control is essential. Some of the major applications include:

Robotics

- Used in robotic arms for accurate joint positioning.
- Useful in pick-and-place robots and automation systems.

CNC Machines

- Stepper motors are a key component in CNC mills and lathes where precise tool positioning is required.

3D Printers

- Stepper motors control the movement of the print head and build platform with high accuracy.

Antenna Positioning Systems

- Enables precise directional alignment of satellite dishes and communication antennas.

Camera Gimbals and Pan-Tilt Systems

- Provides smooth and accurate movement for video recording and surveillance systems.

CONCLUSION

This project successfully demonstrates a low-cost and effective method for achieving **precision angular positioning** using an **Arduino-controlled stepper motor**. The system offers high repeatability and accuracy without the need for complex feedback mechanisms, making it ideal for educational, prototyping, and light industrial purposes.

Through this project, we learned how stepper motors function, how to interface them with Arduino using a motor driver, and how to control angular position using simple programming logic. By adjusting parameters like step angle and microstepping mode, the resolution and smoothness of motion can be further refined.

In conclusion, the combination of Arduino and stepper motor technology provides a powerful tool for precise motion control, opening up numerous possibilities in robotics, automation, and control system design.

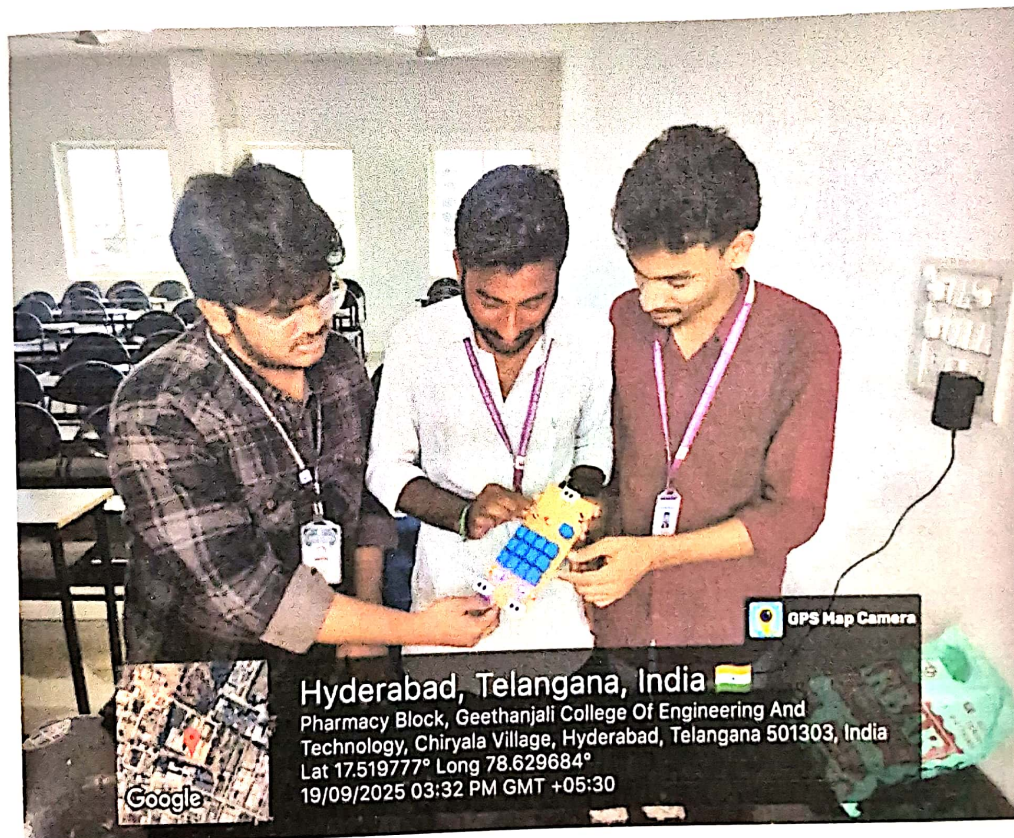
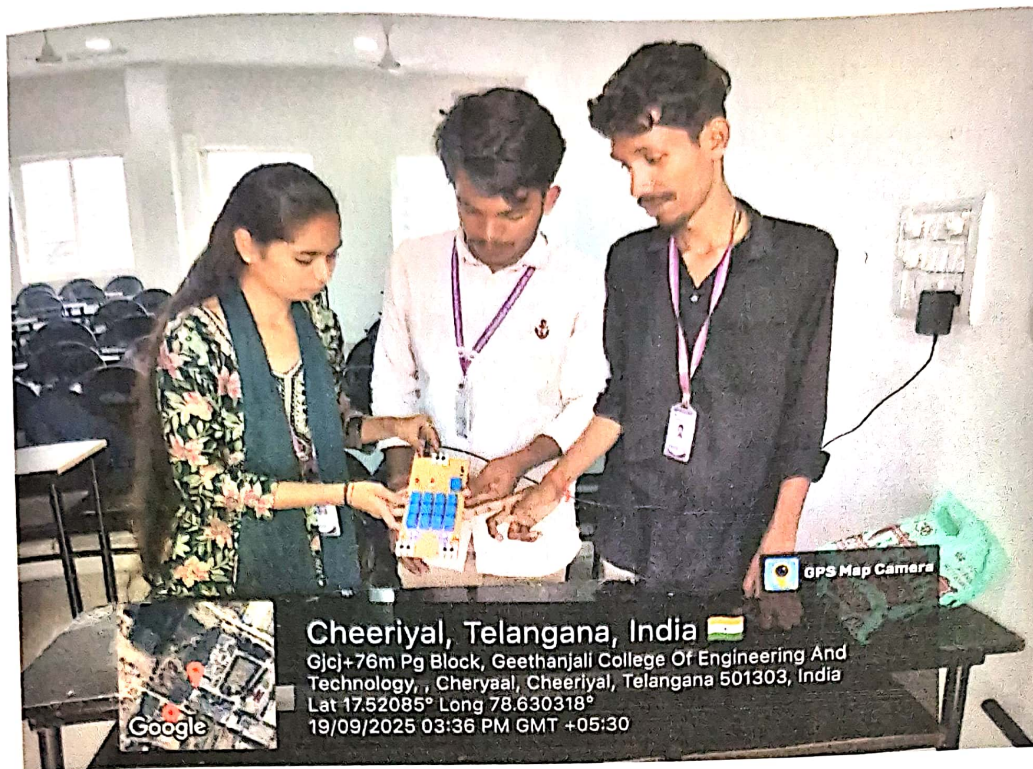
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